

ASSIGNMENT-1

1) Convert the following Boolean function into standard SOP & express in terms of min terms.

$$Q) Y(A, B, C) = AB + A\bar{C} + BC$$

$$= AB(C + \bar{C}) + A\bar{C}(B + \bar{B}) + BC(A + \bar{A})$$

$$= \underline{ABC} + \underline{A\bar{B}\bar{C}} + \underline{AB\bar{C}} + A\bar{B}\bar{C} +$$

$$\underline{ABC} + \bar{A}BC$$

$$= \underset{7}{ABC} + \underset{6}{AB\bar{C}} + \underset{4}{A\bar{B}\bar{C}} + \underset{3}{\bar{A}BC}$$

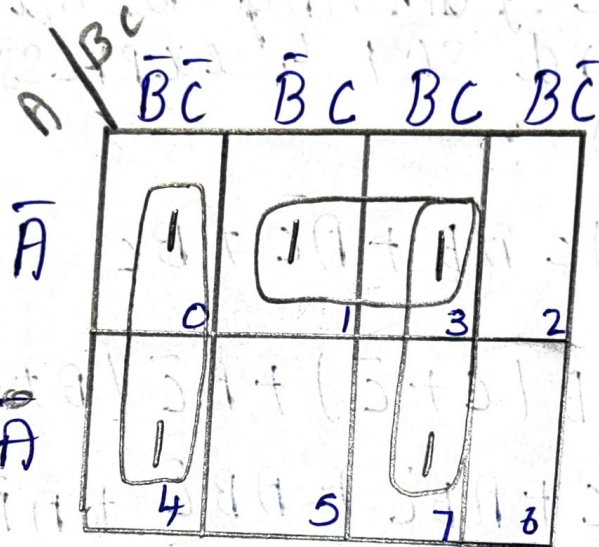
$$F(A, B, C) = \sum m(3, 4, 6, 7)$$

b) $F = A + \hat{B}C$

$$\begin{aligned}
 &= A[B + \bar{B}][C + \bar{C}] + \bar{B}C[A + \bar{A}] \\
 &= A[BC + B\bar{C} + \bar{B}C + \bar{B}\bar{C}] + A\bar{B}C + \bar{A}\bar{B}C \\
 &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C \\
 F &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C = \sum m(1, 4, 5, 6, 7)
 \end{aligned}$$

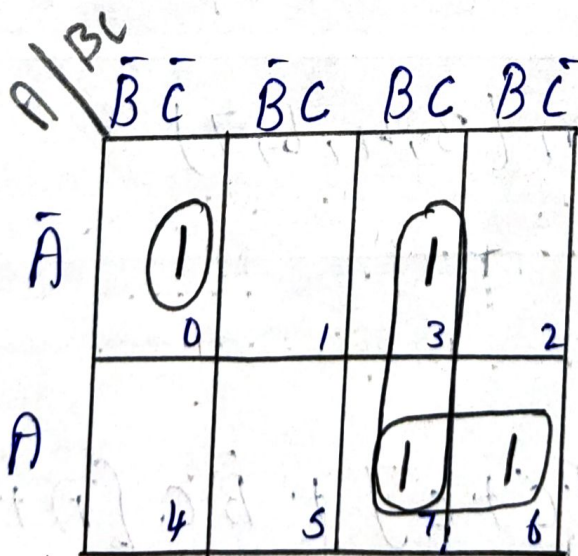
2) Simplify three variable expression using k maps

a) $Y(A, B, C) = \sum m(0, 1, 3, 4, 7)$



$$Y = \bar{B}\bar{C} + \bar{A}C + BC$$

b) $F(A, B, C) = \sum m(0, 3, 6, 7)$



$$= \bar{A}\bar{B}\bar{C} + BC + AB$$

$$c) F(A, B, C) = \Pi M(1, 3, 5, 7)$$

A \ BC		B+C	B+C̄	B̄+C̄	B̄+C
A	Ā	0	1	3	2
	A	4	5	7	6

$$F(A, B, C) = \bar{C}$$

$$d) F(A, B, C) = \Pi M(0, 2, 4, 6)$$

A \ BC		B+C	B+C̄	B̄+C̄	B̄+C
A	Ā	0	1	3	2
	A	4	5	7	6

$$F(A, B, C) = C$$

3) Simplify four variable expression using K maps.

a) $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$

$AB \backslash CD$	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1 0	1 1	1 3	1 2
$\bar{A}B$	1 4	1 5	1 7	1 6
AB	1 12	1 13	1 15	1 14
$A\bar{B}$	1 8	1 9	1 11	1 10

$$F = BD + \bar{B}\bar{D}$$

b) $Y(A, B, C, D) = \sum m(0, 1, 2, 3, 4, 5, 6, 7, 11, 13, 14)$

$AB \backslash CD$	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1 0	1 1	1 3	1 2
$\bar{A}B$	1 4	1 5	1 7	1 6
AB	1 12	1 13	1 15	1 14
$A\bar{B}$	1 8	1 9	1 11	1 10

$$Y = \bar{A} + B\bar{C}D + BC\bar{D} + \bar{B}Cd$$

c) $\chi(A, B, C, D) = \prod M(0, 2, 8, 9, 12, 13, 15)$

$AB \backslash CD$	$C+D$	$C+\bar{D}$	$\bar{C}+\bar{D}$	$\bar{C}+D$
$A+B$	0 0	1	3	0 2
$A+\bar{B}$	4	5	7	6
$\bar{A}+\bar{B}$	0 12	0 13	0 15	14
$\bar{A}+B$	0 8	0 9	10	10

$$\chi = (\bar{A} + C) \cdot (A + B + D) \cdot (\bar{A} + \bar{B} + \bar{D})$$

d) $\chi(A, B, C, D) = \prod M(1, 2, 5, 6, 8, 9, 15)$

$AB \backslash CD$	$C+D$	$C+\bar{D}$	$\bar{C}+\bar{D}$	$\bar{C}+D$
$A+B$	0	0 1	3	0 2
$A+\bar{B}$	4	0 5	7	0 6
$\bar{A}+\bar{B}$	12	13	0 15	14
$\bar{A}+B$	0 8	0 9	10	11

$$\chi = (A + C + D) \cdot (A + \bar{C} + D) \cdot (\bar{A} + B + C) \cdot (\bar{A} + \bar{B} + \bar{C} + \bar{D})$$

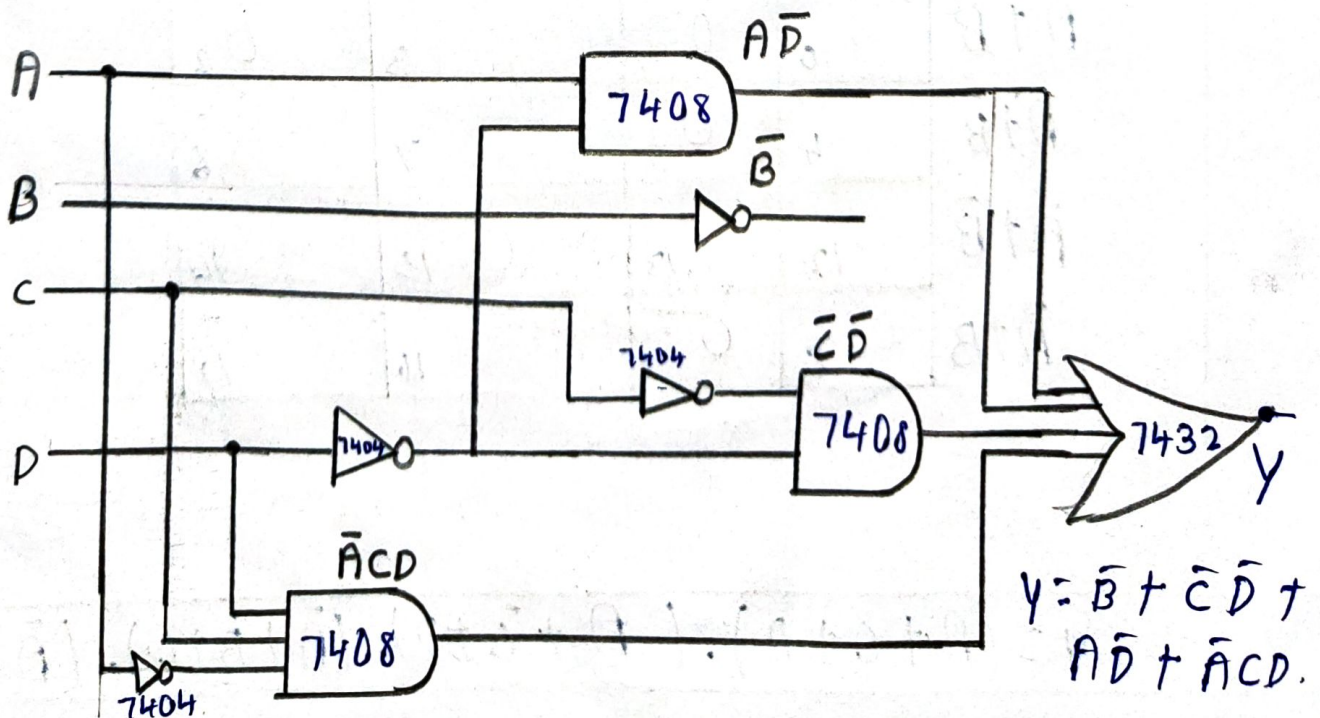
4) Minimize the following using K-map and implement using logic gates.

$$Y = \sum m(0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 14)$$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1 0	1 1	1 2	1 3
$\bar{A}B$	1 4	1 5	1 7	1 6
AB	1 8	1 13	1 15	1 14
$A\bar{B}$	1 8	1 9	1 11	1 10

$$Y = \bar{A}\bar{B} + \bar{A}B + \bar{C}\bar{D} + \bar{A}CD + A\bar{C}\bar{D}$$

$$Y = \bar{B}' + \bar{C}\bar{D} + A\bar{D} + \bar{A}CD$$



5) Minimize the following

a) $Y(A, B, C, D) = \prod M(1, 2, 3, 8, 9, 10, 11, 14) \cdot d(7, 15)$

AB \ CD	CD	C $\bar{D}$	$\bar{C}\bar{D}$	$\bar{C}D$
A $\bar{B}$	0	1	3	2
A $\bar{B}$	4	5	X 7	6
$\bar{A}\bar{B}$	12	13	X 15	14
$\bar{A}\bar{B}$	8	9	11	10

$$Y = (\bar{A} + B) \cdot (\bar{C} + \bar{D}) \cdot (\bar{A} + \bar{C}) \cdot (B + \bar{D})$$

$$Y = (\bar{A} + B) \cdot (B + \bar{C}) \cdot (\bar{A} + \bar{C}) \cdot (B + \bar{D})$$

b) $Y(A, B, C, D) = \sum m(1, 3, 5, 8, 9, 11, 15) + d(2, 13)$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	3	X 2
$\bar{A}\bar{B}$	4	5	7	6
$\bar{A}\bar{B}$	12	X 13	15	14
$\bar{A}\bar{B}$	8	9	11	10

$$= (\bar{B}D) + (\bar{C}D) + (AD) + (A\bar{B}\bar{C})$$

6) FIVE VARIABLE K map

a) $Y(A, B, C, D, E) = \sum m(0, 1, 5, 6, 9, 13, 14, 17, 21, 22, 25, 29)$

BC \ DE		$\bar{D}\bar{E}$	$\bar{D}E$	DE	$D\bar{E}$
		$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{B}\bar{C}$	$\bar{A}\bar{B}$	1	1	3	2
$\bar{B}C$	$\bar{A}B$	4	1	7	1
BC	AB	12	1	15	1
$B\bar{C}$	$A\bar{B}$	8	1	11	10

BC \ DE		$\bar{D}\bar{E}$	$\bar{D}E$	DE	$D\bar{E}$
		$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{B}\bar{C}$	$\bar{A}\bar{B}$	16	1	17	19
$\bar{B}C$	$\bar{A}B$	20	1	21	23
BC	AB	28	1	29	31
$B\bar{C}$	$A\bar{B}$	24	1	25	26

$\bar{A}$

A

$Y = \bar{C}\bar{D} +$

$$Y = \bar{D}E + \bar{A}\bar{B}C\bar{D} + \bar{A}CD\bar{E} + \bar{B}CD\bar{E}$$

common terms - (1, 5, 13, 9) (17, 21, 29, 25)
(6) (22)

b) $Y = \Pi M(3, 4, 7, 11, 15, 19, 21, 23, 27, 29, 28, 31)$

BC/DE		$D+E$	$D+\bar{E}$	$\bar{D}+E$	$\bar{D}+\bar{E}$
$B+C$	$A+B$	0	1	3	2
$B+\bar{C}$	4	5	7	6	
$\bar{B}+\bar{C}$	12	13	15	14	
$\bar{B}+C$	8	9	11	10	

$\bar{A} + A$

BC/DE		$D+E$	$D+\bar{E}$	$\bar{D}+E$	$\bar{D}+\bar{E}$
$B+C$	16	17	19	18	
$B+\bar{C}$	20	21	23	22	
$\bar{B}+\bar{C}$	28	29	31	30	
$\bar{B}+C$	24	25	27	26	

$A + \bar{A}$

$$Y = (\bar{D} + \bar{E}) \cdot (A + B + \bar{C} + D + E).$$

$$(A + \bar{B} + \bar{C} + D) \cdot (\bar{A} + \bar{C} + \bar{E})$$